

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2266

Minnehaha 230 kV
November 30, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2266. The Interconnection Customer proposes a 162.3 MW wind facility interconnecting at the Minnehaha 230 kV substation in the Ohio area. The request was received on December 28, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 751 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,330,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MINNEHAHA-RED BUD 100 KV	STATION S VAN NUYS-MINNEHAHA 230 KV	123.1	135	8.2
RED BUD SUBSTATION-WATERVILLE 100 KV	MINNEHAHA-STA S VAN NUYS 230 KV	114.3	128	8.2
ALLEN JCT-ELMGREN 100 KV L	STA S VAN NUYS-MINNEHAHA 230 KV	107.8	107	10.1
WATERVILLE-NOVEC GARBAR 100 KV line	MINNEHAHA-STATION S VAN NUYS 230 KV	107.5	113	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MINNEHAHA-RED BUD 100 KV Line	Rebuild line with higher-capacity conductor	\$3,072,000	100% (\$3,072,000)
RED BUD SUBSTATION-WATERVILLE 100 KV	Rebuild line with higher-capacity conductor	\$3,623,000	100% (\$3,623,000)
ALLEN JCT-ELMGREN 100 KV L	Reconductor line	\$1,625,000	59% (\$964,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WATERVILLE-NOVEC GARBAR 100 KV line	Reconductor line	\$1,671,000	100% (\$1,671,000)
	Total allocated to project		\$9,330,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.